

# Changelog for GPW WATS 6.01 Connectivity.

Date: 29.03.2024 | Version: GPW0.62

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## 1. DISCLAIMER

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In case of sections of documentation at a High level work progress according to the current version of GPW/WATS Advancement of Documentation, Warsaw Stock Exchange will endeavor to limit changes to these sections of documents to those related to:

1. correcting errors in the documentation or in the software;
2. clarification of the documentation content or removing ambiguity;
3. implementation of approved change requests or;
4. regulatory changes.

## 2. PREFACE

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This document has been prepared by Warsaw Stock Exchange to help in the implementation process of GPW WATS trading platform.

This section describes the basic information about Connectivity. You can learn about the target audience, the document purpose and all the necessary documents you should read in relation to this specification.

### 2.1. TARGET AUDIENCE

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This document has been prepared for development staff, Independent Software Vendors who produce software integrated with GPW WATS, analysts, market participants and all clients who want to deepen their knowledge about GPW WATS.

### 2.2. DOCUMENT PURPOSE

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This document presents all adjustments that has been made in the newest release. The document is always compared with its previous version. Comparison of format changes is not included for readability purposes.

### 2.3. ASSOCIATED DOCUMENTS

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GPW WATS 6.01 Connectivity is a part of GPW WATS documentation set.

Please check the following documents to learn about the construction of Trading System.

- GPW WATS 1.01 Trading System.

Please check the documentation of the trading protocols supported by GPW WATS.

- GPW WATS 2.01 Native Order Gateway Specification
- GPW WATS 2.02 FIX Order Gateway Specification.

Please check the description of the communication with Data Distribution Service.

- GPW WATS 3.01 Market Data Protocol.

Please check the additional documentation, which explains other services provided within GPW WATS.

- GPW WATS 4.01 Drop Copy Gateway
- GPW WATS 4.02 Post Trade Gateway
- GPW WATS 5.01 Risk Management Gateway.

Please check the additional documentation describing the following:

- GPW WATS 2.03 Rejection Codes
- GPW WATS 2.04 BenDec Message Definition Format
- GPW WATS 6.01 Connectivity (this document)

It is recommended that you read the GPW WATS 1.01 Trading System document first.

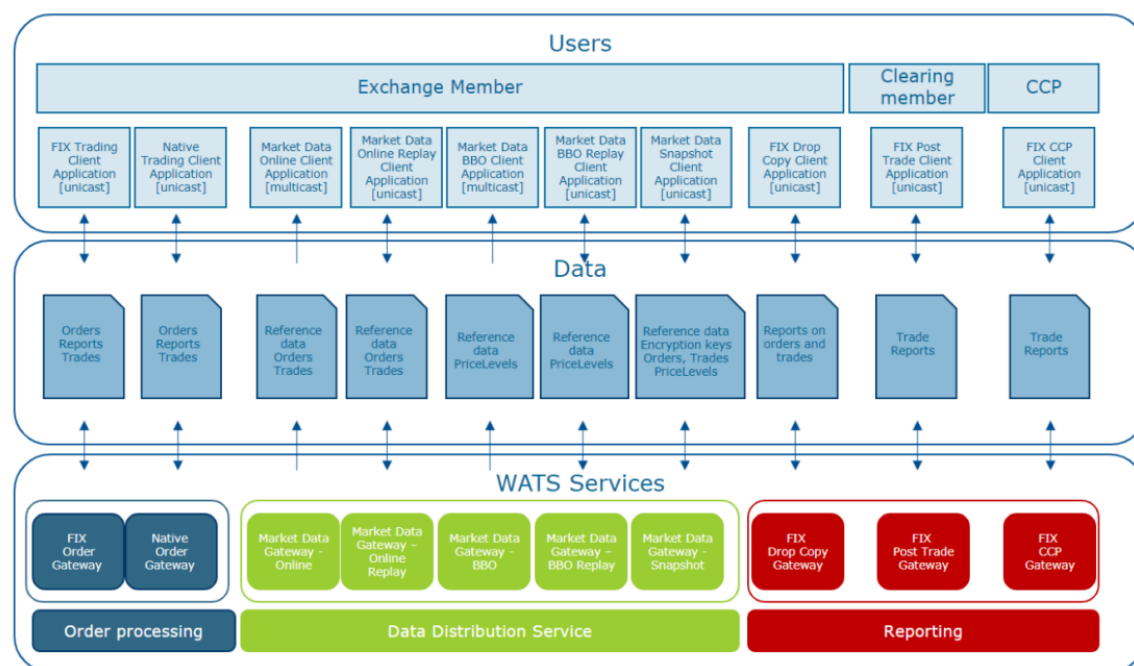
### 3. DOCUMENT HISTORY

| Version       | Date       | Description                                                                                                          |
|---------------|------------|----------------------------------------------------------------------------------------------------------------------|
| <b>0.51</b>   | 29.06.2023 | The initial publication of the documentation.                                                                        |
| <b>0.52</b>   | 26.07.2023 | Publication of v0.52.                                                                                                |
| <b>0.53</b>   | 16.08.2023 | Publication of v0.53.                                                                                                |
| <b>0.54</b>   | 20.09.2023 | Publication of v0.54.                                                                                                |
| <b>0.55</b>   | 5.10.2023  | Two chapters added: <ul style="list-style-type: none"><li>• 6.2.1.2 IPVPN L3</li><li>• 6.2.1.5 HFT Access.</li></ul> |
| <b>0.56</b>   | 08.11.2023 | Publication of v0.56.                                                                                                |
| <b>0.57</b>   | 01.12.2023 | Appendix A has been updated.                                                                                         |
| <b>0.58</b>   | 15.12.2023 | Publication of v0.58.                                                                                                |
| <b>0.59</b>   | 09.01.2024 | Publication of v0.59.                                                                                                |
| <b>0.59.1</b> | 14.02.2024 | Appendix B has been updated.                                                                                         |
| <b>0.62</b>   | 29.03.2024 | IP addresses for the Equinix WA3 have changed.                                                                       |

## 4. GPW WATS SERVICES (INTERFACES)

### 4.1. GPW WATS SERVICES (INTERFACES) – GENERAL OVERVIEW

Figure 1. GPW WATS Services



### 4.2. GPW WATS SERVICES (INTERFACES) – DESCRIPTION

| Services                      | Description                                               |
|-------------------------------|-----------------------------------------------------------|
| OEG::BIN (TCP)                | Order Entry Gateway::BIN                                  |
| OEG::FIX (TCP)                | Order Entry Gateway::FIX                                  |
| OEG::DCP FIX (TCP)            | Order Entry Gateway::Drop copy service (FIX only)         |
| DDS::OMD::Stream (UDP)        | Distribution Data Service - On-line Market Data           |
| DDS::OMD::Replay (TCP)        | Distribution Data Service - On-line Market Data::Replay   |
| DDS::OMD::Snapshot (TCP)      | Distribution Data Service - On-line Market Data::Snapshot |
| DDS::OMD::BBO::Stream (UDP)   | Distribution Data Service - Best Bid Offer                |
| DDS::OMD::BBO::Replay (TCP)   | Distribution Data Service - Best Bid Offer::Replay        |
| DDS::OMD::BBO::Snapshot (TCP) | Distribution Data Service - Best Bid Offer::Snapshot      |

Detailed service specification and functionalities (see: Associated Documents).

### 4.3. ORDER ENTRY GATEWAY (OEG) – ARCHITECTURAL CONTEXT

Order Entry Gateway (OEG) - both types (BIN and FIX) and OEG Drop Copy will be up and running and accessible in both GPW sites (Site A and Site B).

Confidential data will be transmitted over a unicast connection (TCP).



Order Entry Gateways located in Site A in normal operation mode guarantee minimal internal GPW WATS latency between OEG and GPW WATS Core Matching Engine, but total client latency is determined by network (WAN) delays and the routing path "client-to-client".

#### **4.3.1. ORDER ENTRY GATEWAY (OEG::DCP FIX)**

Drop Copy Gateway is a standard functionality which allows receiving private messages for the client.

### **4.4. DATA DISTRIBUTION SYSTEM (DDS) - ARCHITECTURAL CONTEXT**

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DDS::OMD::Stream and its sub-services (Replay and Snapshot) are dedicated to market data transmission.

Market data will be transmitted over a multicast (UDP) simultaneously inside two parallel feeds, both with the same data content (in redundant mode, in different multicast groups: Stream A from Site A only and Stream B from Site B only).

When there is a technical incident or GPW decision, client applications and connectivity configuration should be ready to receive and operate based on single market data feed (Stream A or Stream B) only.

## 5. GPW WATS INSTANCES

### 5.1. TRADING SYSTEM INSTANCES: EUAT, PROD AND PROD-BIS

| Instance                   | Description                                                                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EUAT</b>                | External User Acceptance Test instance<br>Fully functional GPW WATS instance located in two GPW Sites (Site A and Site B).                                                                                  |
| <b>PRE-PROD -&gt; PROD</b> | (planned)<br>PRE-PROD is PROD instance of GPW WATS system before "go-live" located in two GPW Sites (Site A and Site B).                                                                                    |
| <b>PROD-BIS</b>            | (planned)<br>Additional GPW WATS instance installed on minimum set of hardware.<br>Single site instance functionally representative of PRE-PROD (PROD) for client software functional tests and diagnostics |

#### 5.1.1. EUAT SERVICE ORGANISATION

GPW WATS EUAT instances consist of:

- 2 OEG::BIN per Site
- 2 OEG::FIX per Site
- 1 OEG::DCP per Site
- 1 OEG::PTG per Site
- 1 DDS service with four physical Data Feed per Site (see table below):

| Services                              | SITE | Description                                                      | Market/Source |
|---------------------------------------|------|------------------------------------------------------------------|---------------|
| <b>OEG::BIN1(TCP)</b>                 | A    | Order Entry Gateway:: BIN (nb 1)                                 | All           |
| <b>OEG::BIN2(TCP)</b>                 | A    | Order Entry Gateway:: BIN (nb 2)                                 | All           |
| <b>OEG::FIX1 (TCP)</b>                | A    | Order Entry Gateway:: FIX (nb 1)                                 | All           |
| <b>OEG::FIX2 (TCP)</b>                | A    | Order Entry Gateway:: FIX (nb 2)                                 | All           |
| <b>OEG::DCP FIX (TCP)</b>             | A    | Order Entry Gateway:: Drop copy service (FIX only)               | All           |
| <b>OEG::PTG FIX (TCP)</b>             | A    | Order Entry Gateway:: Post Trade Gateway (FIX only)              | All           |
| <b>DDS::OMD::Stream1 (UDP)</b>        | A    | Distribution Data Service1 - On-line Market Data                 | All           |
| <b>DDS::OMD::Replay1 (TCP)</b>        | A    | Distribution Data Service1 - On-line Market Data:: Replay        | All           |
| <b>DDS::OMD::Snapshot1 (TCP)</b>      | A    | Distribution Data Service1 - On-line Market Data:: Snapshot      | All           |
| <b>DDS::OMD::BBO::Stream1 (UDP)</b>   | A    | Distribution Data Service1 - Best Bid Offer                      | All           |
| <b>DDS::OMD::BBO::Replay1 (TCP)</b>   | A    | Distribution Data Service1 - Best Bid Offer:: Replay             | All           |
| <b>DDS::OMD::BBO::Snapshot1 (TCP)</b> | A    | Distribution Data Service1 - Best Bid Offer:: Snapshot           | All           |
| <b>DDS::OMD::Stream2 (UDP)</b>        | A    | Distribution Data Service2 - On-line Market Data                 | No GPW        |
| <b>DDS::OMD::Replay2 (TCP)</b>        | A    | Distribution Data Service2 - On-line Market Data:: Replay        | No GPW        |
| <b>DDS::OMD::Snapshot2 (TCP)</b>      | A    | Distribution Data Service2 - On-line Market Data:: Snapshot      | No GPW        |
| <b>DDS::OMD::BBO::Stream2 (UDP)</b>   | A    | Distribution Data Service2 - Best Bid Offer                      | No GPW        |
| <b>DDS::OMD::BBO::Replay2 (TCP)</b>   | A    | Distribution Data Service2 - Best Bid Offer Market Data:: Replay | No GPW        |



| Services                              | SITE | Description                                                        | Market/Source |
|---------------------------------------|------|--------------------------------------------------------------------|---------------|
| <b>DDS::OMD::BBO::Snapshot2 (TCP)</b> | A    | Distribution Data Service2 - Best Bid Offer Market Data:: Snapshot | No GPW        |

| Services                              | SITE | Description                                                        | Market/Source |
|---------------------------------------|------|--------------------------------------------------------------------|---------------|
| <b>OEG::BIN1(TCP)</b>                 | B    | Order Entry Gateway:: BIN (nb 1)                                   | All           |
| <b>OEG::BIN2(TCP)</b>                 | B    | Order Entry Gateway:: BIN (nb 2)                                   | All           |
| <b>OEG::FIX1 (TCP)</b>                | B    | Order Entry Gateway:: FIX (nb 1)                                   | All           |
| <b>OEG::FIX2 (TCP)</b>                | B    | Order Entry Gateway:: FIX (nb 2)                                   | All           |
| <b>OEG::DCP FIX (TCP)</b>             | B    | Order Entry Gateway:: Drop copy service (FIX only)                 | All           |
| <b>OEG::PTG FIX (TCP)</b>             | B    | Order Entry Gateway:: Post Trade Gateway (FIX only)                | All           |
| <b>DDS::OMD::Stream1 (UDP)</b>        | B    | Distribution Data Service1 - On-line Market Data                   | All           |
| <b>DDS::OMD::Replay1 (TCP)</b>        | B    | Distribution Data Service1 - On-line Market Data:: Replay          | All           |
| <b>DDS::OMD::Snapshot1 (TCP)</b>      | B    | Distribution Data Service1 - On-line Market Data:: Snapshot        | All           |
| <b>DDS::OMD::BBO::Stream1 (UDP)</b>   | B    | Distribution Data Service1 - Best Bid Offer                        | All           |
| <b>DDS::OMD::BBO::Replay1 (TCP)</b>   | B    | Distribution Data Service1 - Best Bid Offer:: Replay               | All           |
| <b>DDS::OMD::BBO::Snapshot1 (TCP)</b> | B    | Distribution Data Service1 - Best Bid Offer:: Snapshot             | All           |
| <b>DDS::OMD::Stream2 (UDP)</b>        | B    | Distribution Data Service2 - On-line Market Data                   | No GPW        |
| <b>DDS::OMD::Replay2 (TCP)</b>        | B    | Distribution Data Service2 - On-line Market Data:: Replay          | No GPW        |
| <b>DDS::OMD::Snapshot2 (TCP)</b>      | B    | Distribution Data Service2 - On-line Market Data:: Snapshot        | No GPW        |
| <b>DDS::OMD::BBO::Stream2 (UDP)</b>   | B    | Distribution Data Service2 - Best Bid Offer                        | No GPW        |
| <b>DDS::OMD::BBO::Replay2 (TCP)</b>   | B    | Distribution Data Service2 - Best Bid Offer Market Data:: Replay   | No GPW        |
| <b>DDS::OMD::BBO::Snapshot2 (TCP)</b> | B    | Distribution Data Service2 - Best Bid Offer Market Data:: Snapshot | No GPW        |

Stream1 – full market data with information from all segments, sources and markets published by GPW WATS.

Stream2 – market data without a GPW feed.

## MARKET DATA ENCRYPTION

Market Data Feed is encrypted.

Certain message fields sent from DDS::OMD::Stream1 and DDS::OMD::Stream2 and correlated DDS::OMD::Replay streams are encrypted. Data decryption requires capturing the encryption key, transmitted in EncryptionKey messages streamed during the initial DDS::OMD::Snapshot session. More details on the decryption process can be found in GPW WATS 3.01 Market Data Gateway Protocol Specification document.

### 5.1.1.1. pre-prod (PROD) SERVICE organization

This will be provided in the next version of the documentation.

### 5.1.1.2. prod-bis SERVICE organization

This will be provided in the next version of the documentation,

### 5.1.2. GPW WATS STANDARD OPERATION MODE

GPW WATS EUAT and PRE-PROD (PROD) system modules (services) operate in parallel in two GPW Data Centers: Site A and Site B.

Based on GPW WATS architecture designs, all hardware units in Site A and Site B can host GPW WATS OEG services.

Under normal operating conditions, the network infrastructure of each GPW Site actively participates in providing redundant connections between GPW and client. Resources (servers) and active trading system modules during standard operation mode in Site A and Site B are up and running.

GPW WATS market data multicast feeds are distributed in parallel: (Stream A) from Site A and (Stream B) from Site B.

By design, DDS Market Data feeds (Stream A and Stream B) are identical. When the network or DDS module malfunctions, the client is obliged to use only one (Stream A or Stream B) Market Data feed accessible from one of the GPW Sites.

### 5.1.3. GPW WATS TECHNICAL EMERGENCY OPERATION MODE

#### 5.1.3.1. Scenario 1 – Client network issue

Connection between the client and GPW WATS excludes SPoF (single point of failure). Routing and network topology should guarantee access to GPW services in the event of a single network connection failure.

Traffic from the client's system should be redirected to an active network connection to GPW WATS platform.

DDS service access and market data feed in this situation is limited to Single stream (Stream A or Stream B) depending on the physical network connectivity termination point.

#### 5.1.3.2. Scenario 2 – GPW WATS single service FAILURE

A Single service can be restarted (rebuilt) on free resources and available for clients without technical connectivity reconfiguration.

#### 5.1.3.3. Scenario 3 – GPW WATS site FAILURE

Services are only available for clients at one specific GPW WATS site.

This scenario does not provide the facility to connect to GPW WATS services configured in an inaccessible site.

#### IMPORTANT:

These scenarios mentioned are not applicable to clients who use the Collocation type of access.

## 6. GPW WATS CONNECTIVITY

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### 6.1. GENERAL RULES

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#### 6.1.1. RESPONSIBILITY AND RECOMMENDATION

GPW does not manage any parts of network or application infrastructure at client's location used to access GPW WATS, thus all client's network infrastructure, in particular edge network devices used to access GPW WATS is managed by a client.

GPW's responsibility for the demarcation point is the network device port managed by telecommunication service providers, which communicate directly with devices managed by GPW. Each Participant is required to order and maintain the lines necessary to communicate with GPW at its own expense. Any operational issues with these lines will be resolved by the Participant and network provider as defined in the SLA.

#### 6.1.2. CLIENT NETWORK DESIGN

The client is responsible for designing the technical infrastructure necessary to implement access to GPW WATS on the client side (including installation and configuration of edge network devices with all aspects of redundancy). However, GPW will provide the general requirements which must be met by each client solution connected to GPW WATS.

#### 6.1.3. CONNECTION SECURITY

Data transfer security between GPW WATS and external client applications is based on the following assumptions:

- WAN connection to GPW WATS is provided over a classified data network.
- It is managed and controlled by individual telecommunication service providers.
- Implementation of connections must comply with GPW requirements. Connections can only be implemented between authorized GPW sites and external client locations.

Data security – Telecommunication service providers provide lines over non-public networks and must ensure adequate separation of communication between clients and only allow communication between clients and GPW. Telecommunication service providers are also responsible for preventing data sniffing and unauthorized access.

#### 6.1.4. GENERAL REMARKS TO CONNECTIVITY AND GPW REQUIREMENTS

The technical documentation published by GPW obliges clients to be compliant with all requirements and recommendations included in it.

Access to GPW services can be realized by clients connected to GPW directly, authorized by GPW network connections. Only telecommunication service providers authorized by GPW can provide connectivity with an appropriate level of redundancy.

GPW is not responsible for any damage caused by improper protection of the application and network resources on the client site (this also applies to telecommunication connections as they are the responsibility and the property of the client).

The client's network system and application connections used for communication with GPW must permit a full diagnosis of its operations. Results of this diagnosis must be made available to all parties involved.

#### **6.1.5. ACTIVE / ACTIVE MODE**

In normal operating mode, both sites (Site A and Site B) are active and accept Client connections.

Both telecommunication lines between the client and Site A and Site B should always be active.

GPW recommends setting the configuration with active OEG and DDS sessions (and connections) in parallel with both GPW sites (Site A and Site B).

## **6.2. CONNECTIVITY – TECHNICAL ASPECTS**

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### **6.2.1. SUPPORTED ACCESS**

#### **6.2.1.1. Equinix Fabric**

GPW WATS will be visible as a dedicated service for Equinix Fabric network users.

Connection standards for Equinix Fabric are defined in the documentation provided by Equinix. The standard connection is based on 100Mbps per client and 10Gbps port. GPW recommends redundancy connection provided by Equinix. Technical standard: fiber optic, single mode (two single fibers).

#### **6.2.1.2. IPVPN L3**

##### **6.2.1.2.1. General Information**

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#### **Redundant DCs**

The GPW WATS system is deployed in two main Equinix sites, WA2 and WA3, which provide service to the system as active/active deployments. Both sites are equipped with the complete technical infrastructure necessary to support trading on the GPW market. Under normal operating conditions, each site's network infrastructure actively provides access to the system.

#### **Redundant connections**

Physical connectivity with both GPW sites (WA2 and WA3) is required to provide full redundant access to GPW WATS. GPW recommendation is to use two different MPLS operators to provide such connectivity to each location.

Both connections must always be active and consider production under normal operation conditions. If any link that the client prefers for unicast or multicast traffic fails, the traffic will automatically switch to the other link using redundancies provided by routing protocols. The above mechanism will ensure an uninterrupted flow of network traffic and reliable access to GPW WATS.

The client application communicating with GPW WATS transaction system should have redundancy and high availability mechanisms supported by the system. In particular, the client application receiving market

data should receive this data from two redundant multicast channels GPW WATS provides over separate paths.

### Redundant market data access

Market data will be available through two redundant multicast channels, A and B streams, sent from Equinix's WA2 and WA3 locations, respectively. These multicast channels will be transmitted via different routes and a completely independent technical infrastructure that is fully redundant.

### Addresses of GPW WATS

The detailed specification of access to GPW WATS, in the particular addresses of external interfaces of the trading system, are provided in **section 6.2.1.2.4 "IP addresses details WA2 and WA3"**, also have been published on the portal dedicated to GPW clients and are available upon request sent to GPW helpdesk: ts@gpw.pl. The addresses of external interfaces of the trading system will be all the required addresses, including those used in failover and disaster recovery procedures. The addresses' usage will be determined in agreement with GPW's telecommunication operator partners.

### Division of responsibilities

The demarcation point of the responsibility of the GPW is the port of the network device administered by the telecommunication operator, which communicates directly with devices administered by the GPW. The GPW will not manage any network and application infrastructure elements on client side. The GPW is unable to monitor the operating status of MPLS operator lines on client side. The client orders and maintains lines necessary to communicate with the GPW at client's own expense. Any operating issues with the lines will be resolved by the client with the MPLS operator as defined in the SLA with the operator. The GPW offers edge application infrastructure (appropriate edge application servers) at its sites to provide access of clients to GPW trading system elements over WAN.

### Security

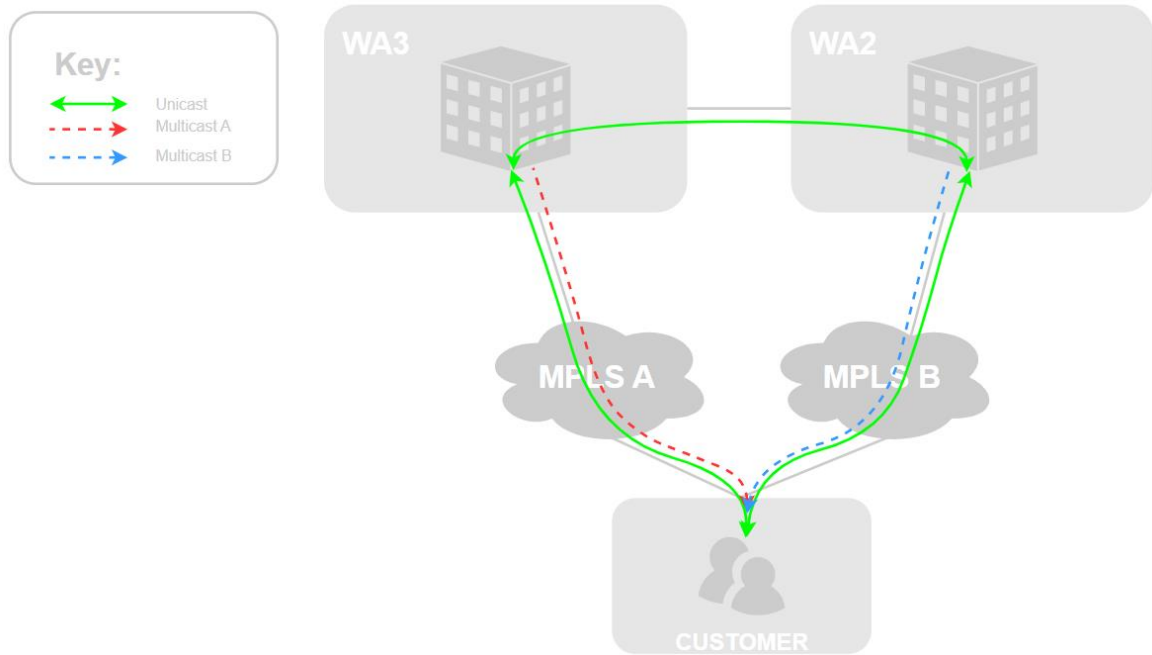
Security of data exchange between the GPW trading system and external client applications is based on the following:

- the wide area network supporting the connection of the GPW trading system with external client applications is not a public network.
- It is managed and controlled by individual telecommunication operators (certified by the GPW).
- The implementation of connections complies with the GPW's requirements; in particular, connections can only be implemented between authorized GPW sites and external client locations.

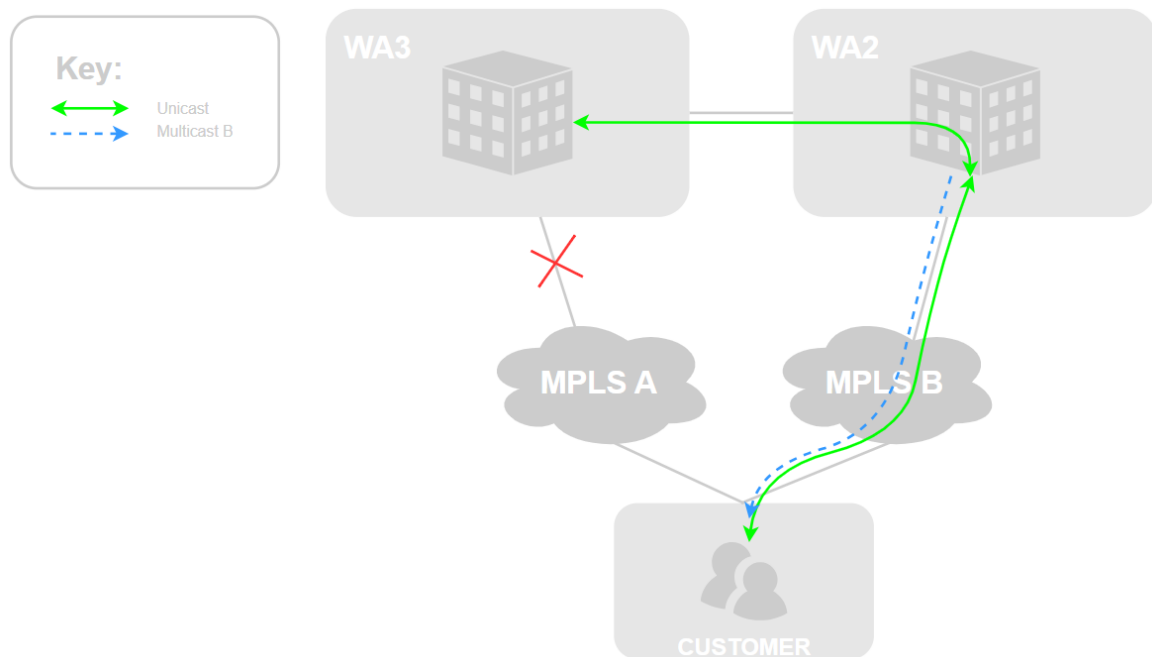
#### 6.2.1.2.2. Typical failover scenarios

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This scenario represents regular operation where both locations are accessible via two independent MPLS providers. Unicast flow is available from both sites. Two multicast separate streams are available from both locations.

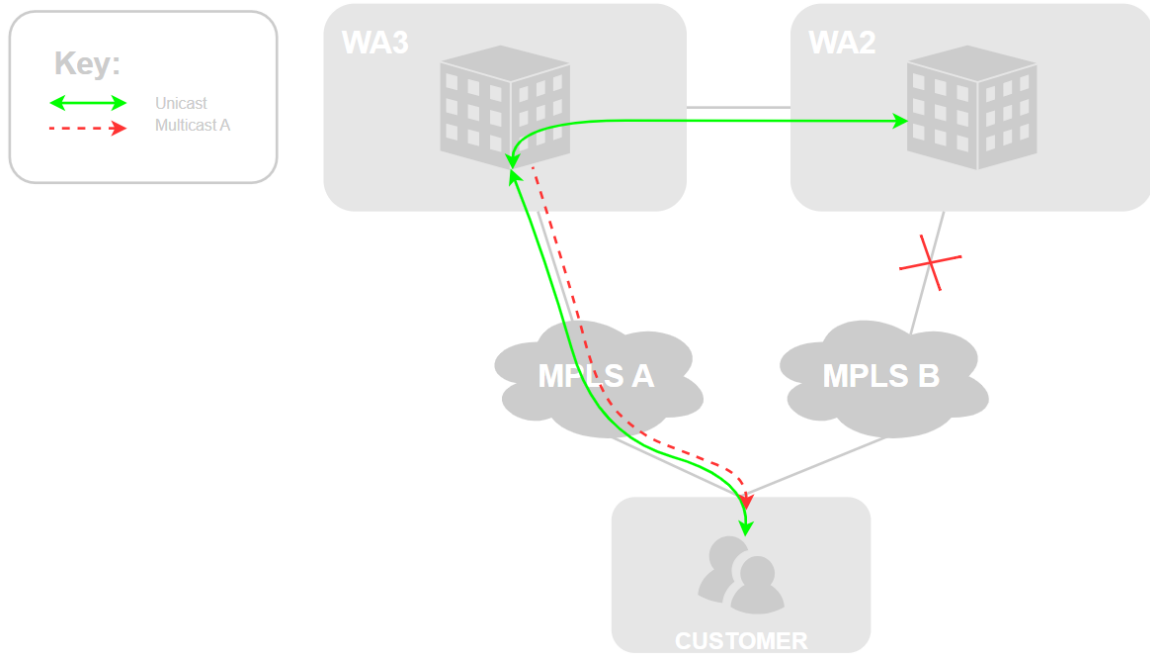


This scenario represents a failure of the MPLS link provider in WA3. Unicast traffic is available from both location WA2 and WA3 and multicast flow is available only from the WA2 site. Based on the application performance or respond time customer should decide which unicast flow is a better choice.

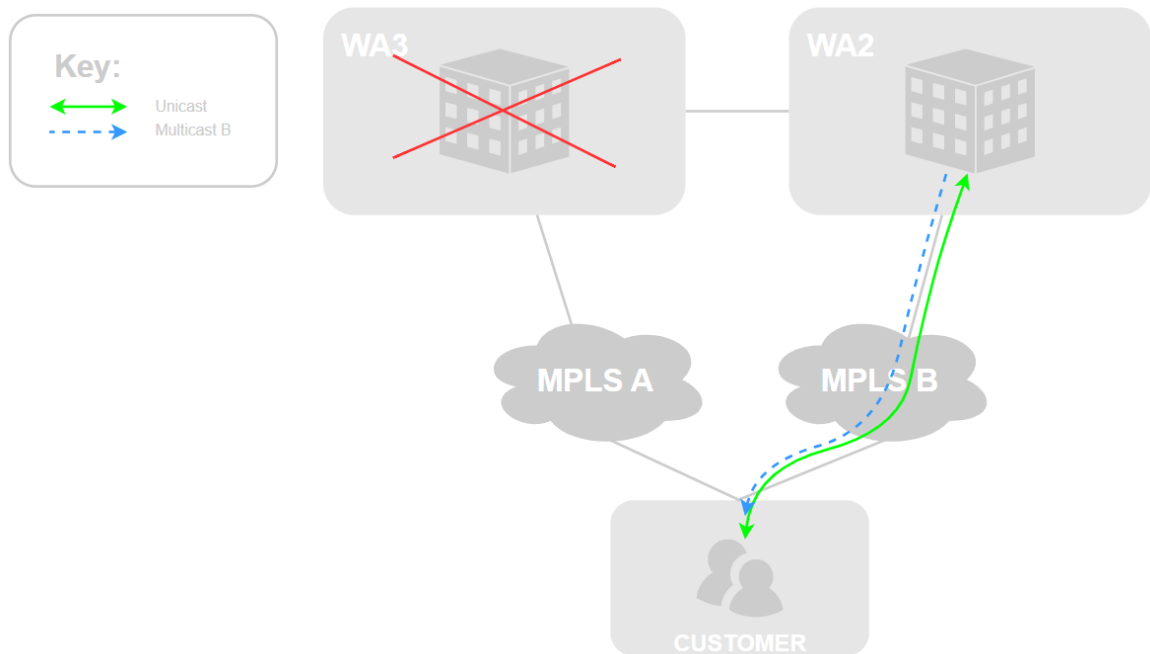


This scenario represents a failure of the MPLS link provider in WA2. Unicast traffic is available from both location WA2 and WA3 and multicast flow is available only from the WA3 site. Based on the application performance or respond time, customers should decide which unicast flow is a better choice.

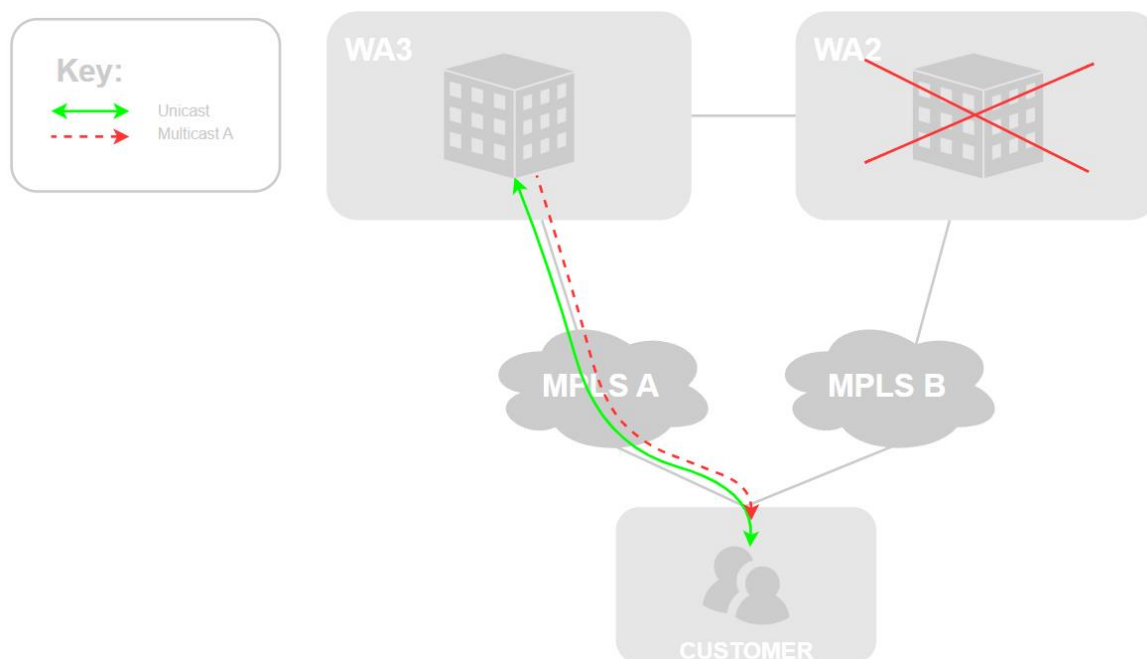




This scenario represents a failure of the WA3 site. Unicast and multicast flows are available only from the WA2 site.



This scenario represents a failure of the WA2 site. Unicast and multicast flows are available only from the WA3 site.



#### 6.2.1.2.3. Telecommunication Operators Supported By GPW

List of Telecommunication Operators supported MPLS connectivity to GPW WATS: see Appendix B.

#### 6.2.1.2.4. Technical Details of The Solution

##### Public (market) data

Public data will be transmitted over a multicast (UDP) and sent simultaneously in two streams, A (FROM WA3) and B (FROM WA2), both with the same data content (in redundant mode, in different multicast groups in stream A and B) in order to enable control of data completeness and cohesion by the Exchange Member, Data Vendor or External Administrator's application and to make retransmit or refresh decisions. Public data under normal conditions as well as under refresh requests of an Exchange Member, Data Vendor or External Administrator will be sent from the GPW over a multicast – communication necessary to send requests (from the Exchange Member, Data Vendor or External Administrator application) to retransmit missing public messages and refresh requests will be sent over a unicast (TCP). Requests to retransmit missing messages from an Exchange Member, Data Vendor or External Administrator to the GPW and retransmission of missing messages from the GPW to the Exchange Member, Data Vendor or External Administrator will be sent over a unicast (TCP). The direct sources of public messages (streams A and B) will be the GPW servers:

- DDS– supporting public data transmission under normal conditions.

Additional services (RTS and snapshot) are unicast.

##### Confidential data (processing orders)

Confidential data will be transmitted over a unicast (TCP); two protocols may be used at application level: Binary and FIX. Communication with the GPW in this regard will be sent to/from OEG (...) edge servers at the GPW.

##### Dynamic routing

The dynamic routing must be available on the border between the Exchange Member, Data Vendor or External Administrators and MPLS operators (unicast border). Implementing dynamic routing between Exchange Members, Data Vendors, External Administrators, and MPLS operators aims to detect network failures and provide dynamic fail-over to other available paths to the GPW WATS application hosted in two GPW Equinix location.

It is important to mention that the GPW offers dynamic BGP routing between its edge devices and every MPLS operator. It is worth noting that there is no direct BGP peering connection between an Exchange Member, Data Vendor, or Third-Party Administrator and the Exchange, regardless of the circumstances.

### Marking packets

The QoS features is used to provide the most desirable flow of traffic through a network. QoS allows to classify the network traffic, police, and prioritize the traffic flow, and help avoid traffic congestion in a network. The control of traffic is based on the fields in the packets that flow through System.

When it comes to outgoing or incoming traffic at the edge network devices of the Exchange Member, Data Vendor or External Administrator, packets need to be marked for traffic prioritization. However, as the GPW does not manage any network devices on the Exchange Member, Data Vendor or External Administrator side, communication with the GPW will be necessary to define the traffic prioritization rules. The packets will be marked in DSCP. The GPW will provide the package marking schema and will take the responsibility of marking the packets in DSCP.

GPW will not mark unicast traffic in QoS from WA2 and WA3. All traffic is treated as best-effort, and to ensure that such traffic provide possible lowest latency and probability of packet loss in the network, customers must discuss their Quality of Service (QoS) configuration with their MPLS service provider, as the provider is responsible for determining how packets are routed and treated on the MPLS network. The provider is also responsible for ensuring the end-to-end quality of the QoS service, which falls outside the scope of GPW.

The production servers' IP addresses are sent individually with TCP ports and other application parameters to each Exchange Member, Data Vendor or External Administrator and agreed at the implementation phase.

### Data security

Ensuring data security is of utmost importance for the GPW. However, it is important to note that the traffic between the GPW and its Exchange Members, Data Vendors or External Administrators is not encrypted.

Telecommunication operators provide lines over non-public networks and must ensure adequate separation of communication between the Exchange Members, Data Vendors or External Administrators and allow communication only between them and the GPW. Additionally, telecommunication operators are responsible for preventing data sniffing and unauthorized access to ensure the security of the data.

### Physical connectivity

Following the recommendation provided by GPW, it is highly advisable to employ SFP+ modules for all physical connections between the GPW edge devices and MPLS providers. It is predominantly because SFP+ transceivers facilitate upgrading speeds to higher than 1G in the future without incurring the need for module replacement.

In addition, single-mode modules should be utilized. Nevertheless, all technical specifications should be taken into consideration during the implementation of the solution.

The physical connection between GPW edge devices and the MPLS provider in each Equinix colocation can be shared by multiple customers using the same MPLS provider and GPW recommends MPLS providers to establish two redundant physical connections, each one can be connected to a cluster of GPW edge devices.

The GPW recommends a 150M bandwidth for multicast and unicast traffic per client. Nevertheless, the MPLS service provider must ensure logical separation between clients when multiple clients from the same provider share a physical connection and ensure proper propagation of GPW prefixes to the client's MPLS network.

Another option is to connect the operator's MPLS edge device via two physical links bundled into a logical port-channel to two separate WATS switches utilizing MLAG technology to provide better resilience on the WATS side. It is recommended that this option be discussed with an operator during the initial conversation regarding service provisioning in WA2 or WA3 locations.

**GPW recommends full redundancy on each level of client infrastructure for GPW WATS services and separation from currently used network devices.**

#### IP addresses details WA2 and WA3

As mentioned above the BGP is the preferred routing protocol for the logical connection between the GPW and MPLS provider. As part of the project, GPW has reserved the public address 91.234.144.0/24 and the BGP AS number **24732** for peer-to-peer communication with all MPLS providers who will provide the GPW WATS service to their customers. Location WA2 will use the range **91.234.144.0/25**, and location WA3 will use the range **91.234.144.128/25**.

For BGP sessions, it is recommended to use BFD for sub-second failover in case of a physical link or hardware issue. BFD provides much faster failure detection than standard BGP hold time, even though BGP timers can be adjusted to low values. Additionally, it is highly advised to use MD5 authentication to secure BGP TCP sessions between peers.

The following table shows the address breakdown for the Equinix WA2.

| WA2                     |                              |                           |
|-------------------------|------------------------------|---------------------------|
| Subnet                  | Usable IP                    | Comments                  |
| <b>91.234.144.1/32</b>  | 91.234.144.1- 91.234.144.1   | GPW WATS multicast PIM RP |
| <b>91.234.144.32/28</b> | 91.234.144.33- 91.234.144.46 | GPW WATS multicast source |
| <b>91.234.144.64/27</b> | 91.234.144.65- 91.234.144.94 | GPW WATS unicast servers  |

The only IP address **91.234.144.1** will be used on GPW side for multicast PIM RP in the WA2 location, with Anycast-RP providing redundancy.

Each MPLS P2P connection will be allocated a dedicated /30 subnet. Further details regarding the allocation will be determined during the implementation stage.

In certain cases, if the MPLS operator wants to establish a greater level of redundancy and connect their device to GPW edge devices in WA2 via port-channel technology, a dedicated /29 subnet will be allocated.

Table below shows allocated range for WA2 MPLS P2P connections.

| WA2    |          |
|--------|----------|
| Subnet | Comments |

| WA2                 |                      |
|---------------------|----------------------|
| <b>10.52.4.0/24</b> | MPLS P2P connections |
| <b>10.52.5.0/24</b> | MPLS P2P connections |

The following table shows the address breakdown for the Equinix WA3.

| WA3                      |                                |                           |
|--------------------------|--------------------------------|---------------------------|
| Subnet                   | Usable IP                      | Comments                  |
| <b>91.234.144.129/32</b> | 91.234.144.129- 91.234.144.129 | GPW WATS multicast PIM RP |
| <b>91.234.144.160/28</b> | 91.234.144.161- 91.234.144.174 | GPW WATS multicast source |
| <b>91.234.144.192/27</b> | 91.234.144.193- 91.234.144.222 | GPW WATS unicast servers  |

The only IP address **91.234.144.129** will be used on the GPW side for multicast PIM RP in the WA3 location, with Anycast-RP providing redundancy.

Each MPLS P2P connection will be allocated a dedicated /30 subnet. Further details regarding the allocation will be determined during the implementation stage.

In certain cases, if the MPLS operator wants to establish a greater level of redundancy and connect their device to GPW edge devices in WA3 via port-channel technology, a dedicated /29 subnet will be allocated.

Table below shows allocated range for WA3 MPLS P2P connections.

| WA3                 |                      |
|---------------------|----------------------|
| Subnet              | Comments             |
| <b>10.56.4.0/24</b> | MPLS P2P connections |
| <b>10.56.5.0/24</b> | MPLS P2P connections |

In addition to the above IP addresses for the WA2 and WA3 Equinix locations, GPW has allocated a dedicated private address space **10.222.0./16** and **10.226.0.0/16**, respectively, from which customers will be allocated small subnets for a dedicated GPW WATS environment built on the customer side. That subnet will be advertised by customer BGP session to the MPLS provider and propagated to the WATS environment on the GPW side. The GPW's imposition of addressing for WATS environments on the client side will help avoid situations with overlapping IP addresses and the use of NAT, which will simplify deployment.

Two tables below show the private range allocation for customers GPW WATS environment.

| WA2                  |                                                            |
|----------------------|------------------------------------------------------------|
| Subnet               | Comments                                                   |
| <b>10.222.0.0/16</b> | Private range allocated for customers GPW WATS environment |

| WA3                  |                                                            |
|----------------------|------------------------------------------------------------|
| Subnet               | Comments                                                   |
| <b>10.226.0.0/16</b> | Private range allocated for customers GPW WATS environment |

Exact subnet allocation will be decided during the implantation phase and validated case by case with customers.

#### **6.2.1.3. Leased lines**

Leased lines are point-to-point Ethernet lines between the client location and GPW Sites. GPW allows installation of physical access lines used for communication with GPW only over professional fiber optic cables, which ensure the necessary quality, scalability, and security of data transmission. It is expected that the client orders connections with the necessary capacity from two different telecommunication service providers, one to GPW Site A and one to GPW Site B, which will ensure redundancy in communication with GPW WATS system. The network capacity (bandwidth) is identical for both sites.

Leased line connection to GPW WATS can be set up in cooperation with any telecommunication service provider that will be able to provide a professional fiber optic circuit from a client location to GPW Sites over their own infrastructure or using third party shared infrastructure (not limited to the IPVPN L3 Telecom Provider list).

#### **6.2.1.4. VPN Connection**

VPN connections guarantee access to OEG services only.

DDS: OMD and DDS::BBO services will not be accessible via VPN connection.

#### **6.2.1.5. HFT Access**

##### **6.2.1.5.1. General Information**

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#### **Redundant DCs**

The GPW WATS system is deployed in two main Equinix sites, WA2 and WA3, which provide service to the system as active/active deployments. Both sites are equipped with the complete technical infrastructure necessary to support trading on the GPW market. Under normal operating conditions, each site's network infrastructure actively provides access to the system.

#### **Redundant connections**

Physical connectivity to both GPW locations (WA2 and WA3) is required to ensure fully redundant access to the GPW WATS system. GPW recommends using two different physical cables that will be connected to two switches in each location to provide maximum redundancy.

Both connections must always be active and consider production under normal operation conditions. If any location that the client prefers for unicast or multicast traffic fails, the traffic will automatically switch to the other location using redundancies provided by routing protocols. The above mechanism will ensure an uninterrupted flow of network traffic and reliable access to the GPW WATS system.

The client application communicating with the GPW WATS transaction system should have redundancy and high availability mechanisms supported by the system. In particular, the client application receiving market data should receive this data from two redundant multicast channels the GPW WATS system provides over separate paths.

#### **Redundant market data access**

Market data will be available through two redundant multicast channels, A and B streams, sent from Equinix's WA2 and WA3 locations, respectively. These multicast channels will be transmitted via different routes and a completely independent technical infrastructure that is fully redundant.

### Addresses of the GPW WATS system

The detailed specification of access to the GPW WATS system, in particular the addresses of external interfaces of the trading system, are provided in **section 6.2.1.5.3 "IP addresses details WA2 and WA3"**, also have been published on the portal dedicated to GPW clients and are available upon request sent to GPW helpdesk: ts@gpw.pl. The addresses of external interfaces of the trading system will be all the required addresses, including those used in failover and disaster recovery procedures. The addresses' usage will be determined in agreement with GPW's telecommunication operator partners.

### Division of responsibilities

The demarcation point of the responsibility of the GPW is the port of the network device manage by the customer, which is directly connected with devices administered by the GPW. The GPW will not manage any network and application infrastructure elements on client side. The GPW can monitor the status of direct links between GPW edge and customer, however usually inter-connections lines are provided by Equinix and any operating issues with the lines will be resolved by the client with the Equinix operator as defined in the SLA with the operator. The GPW offers edge application infrastructure (appropriate edge application servers) at its sites to provide access of clients to GPW trading system elements over direct link connected to GPW infrastructure.

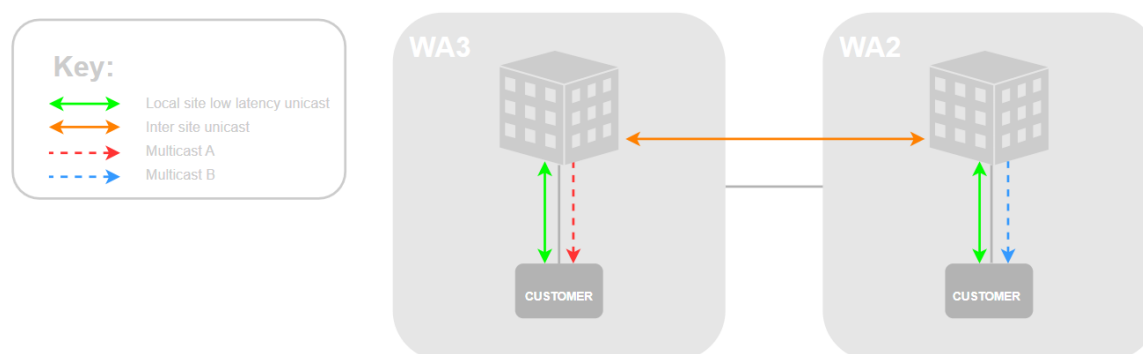
### Security

Security of data exchange between the GPW trading system and external client applications is based on the following:

- Direct connection of the GPW trading system with external client applications is based on a private dedicated physical link.
- It is managed and controlled by WSA and external customer only.
- The implementation of connections complies with the GPW's requirements; in particular, connections is a direct physical cable between two entities only.

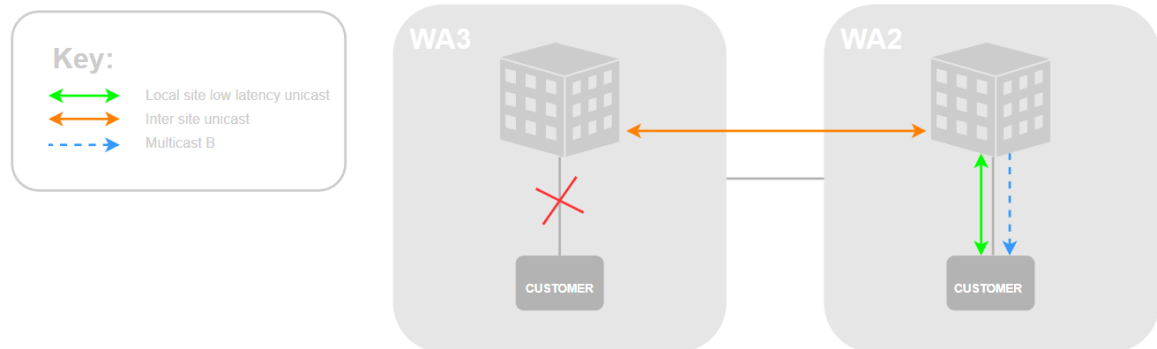
#### 6.2.1.5.2. Typical Failover Scenarios

This scenario represents regular operation where both locations are accessible via two independent private connections in each Equinix colocation. The low latency unicast flow is available from each site. Two multicast separate streams are available from both locations.

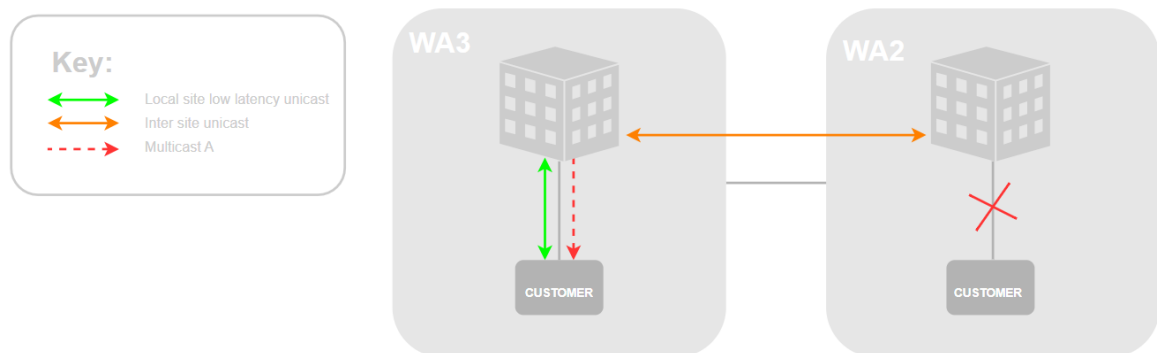




This scenario represents a failure of the direct connection in WA3. The low latency unicast traffic is available from a location WA2, and not low latency unicast traffic is available from WA3 over DCI link. Multicast flow is available only from the WA2 site.



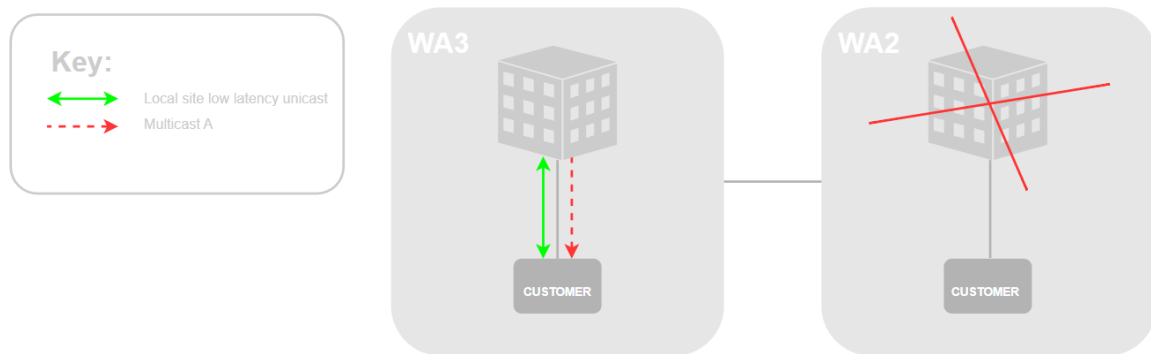
This scenario represents a failure of the direct connection in WA2. The low latency unicast traffic is available from a location WA3, and not low latency unicast traffic is available from WA2 over DCI link. Multicast flow is available only from the WA3 site.



This scenario represents a failure of the WA3 site. Unicast and multicast flows are available only from the WA2 site.



This scenario represents a failure of the WA2 site. Unicast and multicast flows are available only from the WA3 site.



### 6.2.1.5.3. Technical Details of The Solution

#### Public (market) data

Public data will be transmitted over a multicast (UDP) and sent simultaneously in two streams, A (FROM WA3) and B (FROM WA2), both with the same data content (in redundant mode, in different multicast groups in stream A and B) in order to enable control of data completeness and cohesion by the Exchange Member, Data Vendor or External Administrator's application and to make retransmit or refresh decisions. Public data under normal conditions as well as under refresh requests of an Exchange Member, Data Vendor or External Administrator will be sent from the GPW over a multicast – communication necessary to send requests (from the Exchange Member, Data Vendor or External Administrator application) to retransmit missing public messages and refresh requests will be sent over a unicast (TCP). Requests to retransmit missing messages from an Exchange Member, Data Vendor or External Administrator to the GPW and retransmission of missing messages from the GPW to the Exchange Member, Data Vendor or External Administrator will be sent over a unicast (TCP). The direct sources of public messages (streams A and B) will be the GPW servers:

- DDS – supporting public data transmission under normal conditions.

Additional services (RTS and SnapShot) are unicast.

#### Confidential data (processing orders)

Confidential data will be transmitted over a unicast (TCP); two protocols may be used at application level: Binary and FIX. Communication with the GPW in this regard will be sent to/from OEG (...) edge servers at the GPW.

#### Dynamic routing

The recommended dynamic routing protocol between WATS FTD edge and customer edge is a BGP for unicast and PIM-SM for multicast. Implementing dynamic routing between both entities aims to detect network failures and provide dynamic fail-over to other available paths to the GPW WATS application hosted in two GPW Equinix location.

WSA strongly recommended using the BFD feature to provide sub seconds failover for any control plane issues, which can speed up failure detection much faster than relying on setting BGP or PIM timers.

#### Marking packets

The QoS features is used to provide the most desirable flow of traffic through a network. QoS allows to classify the network traffic, police, and prioritize the traffic flow, and help avoid traffic congestion in a network. The control of traffic is based on the fields in the packets that flow through the system.

When it comes to outgoing or incoming traffic at the edge network devices, packets need to be marked for traffic prioritization. However, as the GPW does not manage any network devices on the customer side, communication with the GPW will be necessary to define the traffic prioritization rules. The packets will be marked in DSCP. The GPW will provide the package marking schema and will take the responsibility of marking the packets in DSCP.

GPW will mark all unicast traffic from WA2 and WA3 subnets **91.234.144.32/27** and **91.234.144.160/27** respectively as EF to ensure that such traffic is always treated as priority traffic and to ensure that such traffic provide possible lowest latency and probability of packet loss in the network. Customers must ensure that such packets are treated appropriately on their network, as it is the customer's responsibility to determine how packets are routed and treated within its network.

The production servers' IP addresses are sent individually with TCP ports and other application parameters to each customer and agreed at the implementation phase.

### Data security

Ensuring data security is of utmost importance for the GPW. However, it is important to note that the traffic between the GPW and its customers is not encrypted.

### Physical connectivity

Following the recommendation provided by GPW, it is highly advisable to employ SFP+ modules for all physical connections between the GPW edge devices and customer edge. It is predominantly because SFP+ transceivers facilitate upgrading speeds to higher than 1G in the future without incurring the need for module replacement.

In addition, single-mode modules should be utilized. Nevertheless, all technical specifications should be taken into consideration during the implementation of the solution.

WATS recommends connecting the customer edge device via two physical links bundled into a logical port-channel to two separate WATS switches utilizing MLAG technology to provide better resilience on the WATS side. It is recommended that this option be discussed with a customer during the initial conversation regarding service provisioning in WA2 or WA3 locations.

### IP addresses details WA2 and WA3

As mentioned above, the BGP is the preferred routing protocol for the logical connection between the GPW and customer edge. As part of the project, GPW has reserved the public address 91.234.144.0/24 and the BGP AS number **24732** for peer-to-peer communication with all customers who want to leverage GPW WATS service. Location WA2 will use the range **91.234.144.0/25**, and location WA3 will use the range **91.234.144.128/25**.

For BGP sessions, it is recommended to use BFD for sub second failover in case of a physical link or hardware issue. BFD provides much faster failure detection than standard BGP hold time, even though BGP timers can be adjusted to low values. Additionally, it is highly advised to use MD5 authentication to secure BGP TCP sessions between peers.

The following table shows the address breakdown for the Equinix WA2.

| WA2                     |                              |                           |
|-------------------------|------------------------------|---------------------------|
| Subnet                  | Usable IP                    | Comments                  |
| <b>91.234.144.0/30</b>  | 91.234.144.1- 91.234.144.2   | GPW WATS multicast PIM RP |
| <b>91.234.144.8/29</b>  | 91.234.144.9- 91.234.144.14  | GPW WATS multicast source |
| <b>91.234.144.32/27</b> | 91.234.144.32- 91.234.144.62 | GPW WATS unicast servers  |

The only IP address **91.234.144.1** will be used on the GPW side for multicast PIM RP in the WA2 location, with Anycast-RP providing redundancy.

Each customer connection will be allocated a dedicated /29 subnet. Further details regarding the allocation will be determined during the implementation stage.

Table below shows allocated range for WA2 private transit connections.

| WA2                 |                             |
|---------------------|-----------------------------|
| Subnet              | Comments                    |
| <b>10.52.6.0/24</b> | Private transit connections |
| <b>10.52.7.0/24</b> | Private transit connections |

The following table shows the address breakdown for the Equinix WA3.

| WA3                      |                               |                           |
|--------------------------|-------------------------------|---------------------------|
| Subnet                   | Usable IP                     | Comments                  |
| <b>91.234.144.128/30</b> | 91.234.144.129-91.234.144.130 | GPW WATS multicast PIM RP |
| <b>91.234.144.136/29</b> | 91.234.144.137-91.234.144.142 | GPW WATS multicast source |
| <b>91.234.144.160/27</b> | 91.234.144.161-91.234.144.190 | GPW WATS unicast servers  |

The only IP address **91.234.144.129** will be used on the GPW side for multicast PIM RP in the WA3 location, with Anycast-RP providing redundancy.

Each customer connection will be allocated a dedicated /29 subnet. Further details regarding the allocation will be determined during the implementation stage.

Table below shows allocated range for WA3 private transit connections.

| WA3                 |                             |
|---------------------|-----------------------------|
| Subnet              | Comments                    |
| <b>10.56.6.0/24</b> | Private transit connections |
| <b>10.56.7.0/24</b> | Private transit connections |

In addition to the above IP addresses for the WA2 and WA3 Equinix locations, GPW has allocated a dedicated private address space **10.222.0.0/16** and **10.226.0.0/16**, respectively, from which customers will be allocated small subnets for a dedicated GPW WATS environment built on the customer side. That subnet will be advertised by customer BGP session directly to the WATS environment on the GPW side. The GPW's imposition of addressing for WATS environments on the client side will help avoid situations with overlapping IP addresses and the use of NAT, which will simplify deployment.

Two tables below show the private range allocation for customers GPW WATS environment.

| WA2           |                                                            |
|---------------|------------------------------------------------------------|
| Subnet        | Comments                                                   |
| 10.222.0.0/16 | Private range allocated for customers GPW WATS environment |

| WA3           |                                                            |
|---------------|------------------------------------------------------------|
| Subnet        | Comments                                                   |
| 10.226.0.0/16 | Private range allocated for customers GPW WATS environment |

Exact subnet allocation will be decided during the implantation phase and validated case by case with customers.

### 6.2.2. TELECOMMUNICATION SERVICES PROVIDERS

For IPVPN L3 connectivity, GPW currently supports and authorizes telecommunication service providers listed in Appendix B.

GPW certified telecommunication service providers are obliged to require GPW Clients to obtain the formal GPW approval of each connection to be set up from the client's location to GPW Sites. The requirement includes parameters of inter-connections between GPW Participant's networks and telecommunication service providers.

Currently, in cooperation with telecommunication service providers, GPW is planning and preparing the network infrastructure needed to distribute GPW WATS services. More detailed information and a detailed access procedure will be provided in later versions of this document.

### 6.2.3. ADDITIONAL INFORMATION

Leased lines connectivity to GPW WATS can be set up by any telecommunication service providers depending on the client's business needs.

### 6.2.4. COLOCATION SERVICE

Colocation is a dedicated service at Site A. All colocation clients have guaranteed access to GPW WATS services (DDS and OEG) with identical latency.

All technical aspects of client equipment installation should be confirmed with Equinix directly.

Connectivity parameters for collation: see Appendix A.

## 6.3. CONNECTIVITY DETAILS

### 6.3.1. GPW LOCATIONS (DATA CENTERS)

GPW WATS is in two locations: Site A and Site B. Both locations are equipped with the complete technical infrastructure necessary to support trading on GPW. In normal mode, network infrastructure of both locations actively participates in redundant access to the processing center, where the active Core Matching Engine Module is started.

Core Matching Engine Module location and co-location service (in the future) determine Site A as a privileged location.

GPW recommended connectivity technical standards.

#### 6.3.1.1. Network traffic routing

IP communication between GPW edge network devices and clients will be based on BGP routing. As for IPVPN L3 technology, GPW will have BGP peering with all certified telecommunication providers at each site providing GPW WATS services to clients. For leased lines GPW will have direct BGP peering with the client edge devices.

#### 6.3.1.2. Client addressing

As for IPVPN L3 connections, the IP addresses of clients and network parameters are agreed upon between GPW and telecommunication service providers per the request of GPW Client. For leased line connections, client connectivity details are discussed directly with GPW. All connectivity details (peering and source addressing, AS number) are discussed with GPW and telecommunication service providers or clients before establishing connectivity to GPW. GPW will provide GPW WATS services from both GPW Sites using public IPv4 addressing and an AS number.

#### 6.3.1.3. Unicast communication

Unicast communication between GPW and client is based on BGP routing. GPW WATS Unicast services provided from GPW locations will be presented as IPv4 BGP prefixes from both GPW Sites.

When there is a network failure, connection between the client and GPW WATS unicast services, considered the main unicast traffic, should be switched to an alternative network path based on BGP routing.

#### 6.3.1.4. Multicast communication

Multicast communication is based on BGP routing and the PIM protocol. Based on GPW WATS Data Distribution System architecture, connectivity to the market data feed should be as follows: Multicast A from Site A should be configured as a separate connection to the market data feed, Multicast B from Site B. In normal operating mode, Multicast A and Multicast B will be distributed from two different GPW sites. Due to bandwidth restrictions, during network failure, only one market data feed should be received and routing both market data feeds using one network link is not allowed.

#### 6.3.1.5. Bandwidth

Bandwidth specification – target GPW recommendation for GO-LIVE date (see table below).

NOTICE: Recommendation based on:

- Contractual WATS performance (50.000 orders/sec)

| Service (one Site)                                             | Bandwidth EUAT (One Site) | Bandwidth PrePROD (One Site) | Bandwidth PROD-BIS (Site A only) |
|----------------------------------------------------------------|---------------------------|------------------------------|----------------------------------|
| OEG::BIN (for all FIX and BIN session till 50.000 order / sec) | 10 Mbps                   | 10 Mbps                      | 10 Mbps                          |
| OEG::FIX (see above)                                           | 10 Mbps                   | 10 Mbps                      | 10 Mbps                          |
| DDS::OMD::Stream1                                              | 40 Mbps                   | 40 Mbps                      | 40 Mbps                          |
| DDS::OMD::Stream2                                              | 10 Mbps                   | 10 Mbps                      | n/a                              |

| Service (one Site)                                                                                   | Bandwidth<br>EUAT<br>(One Site) | Bandwidth<br>PrePROD<br>(One Site) | Bandwidth<br>PROD-BIS<br>(Site A only) |
|------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------|----------------------------------------|
| DDS::BBO::Stream1                                                                                    | 40 Mbps                         | 40 Mbps                            | 40 Mbps                                |
| DDS::BBO::Stream2                                                                                    | 10 Mbps                         | 10 Mbps                            | n/a                                    |
| OEG::BIN (or FIX) + DDS::OMD::Stream1                                                                | 50 Mbps                         | 50 Mbps                            | 50 Mbps                                |
| OEG::BIN (or FIX) + DDS::OMD::Stream2                                                                | 20 Mbps                         | 20 Mbps                            | n/a                                    |
| OEG::BIN (or FIX) + DDS::BBO::Stream1                                                                | 50 Mbps                         | 50 Mbps                            | 50 Mbps                                |
| OEG::BIN (or FIX) + DDS::BBO::Stream2                                                                | 20 Mbps                         | 20 Mbps                            | n/a                                    |
| OEG::BIN (or FIX) + DDS::OMD::Stream1 + DDS::BBO::Stream1                                            | 100 Mbps                        | 100 Mbps                           | n/a                                    |
| OEG::BIN (or FIX) + DDS::OMD::Stream1 + DDS::BBO::Stream1<br>+ DDS::OMD::Stream2 + DDS::BBO::Stream2 | 200 Mbps                        | 200 Mbps                           | n/a                                    |

Stream1 – full market data with information from all segments, sources, and market operators.

Stream2 – market data without the GPW market data feed.

For GPW WATS, quality of Service (QOS DSCP) mechanisms will be used, as well as the marking of packets in traffic classes, to prevent different types of traffic competing for the same bandwidth. GPW WATS QOS parameters must be transferred from GPW to clients.

Clients should consider the possibility of reserving additional bandwidth above the value required for EUAT/PRE-PROD (PROD). Exact bandwidth requirements may be different and depend on client needs, including private and public data, total traffic, and instances of GPW WATS running on the same link.

#### IMPORTANT:

Due to the planned performance tests, GPW on the EUAT environment until opening the PRE-PROD GPW WATS instance, GPW recommends:

- When Client decides to connect to DDS and OEG services – GPW recommends allocating **100 Mbps**.
- When Client decides not to connect to DDS service and connect only to OEG service – GPW recommends allocating **at least 10 Mbps**.



## 7. GPW WATS CONNECTIVITY PROCEDURES

### 7.1. GPW WATS EUAT CONNECTIVITY GUIDE

Clients who would like to connect to GPW WATS EUAT are kindly asked for their cooperation in taking the necessary preparation steps described below.

#### 7.1.1. STEPS TO CONNECT

Connecting to EUAT GPW WATS requires taking a series of actions:

- Readyng software implementing protocols supported by the system.
- Obtaining appropriate IP addresses of GPW WATS services.
- Obtaining credentials, required to authorize protocol login procedure to the system (i.e., connection ID and token for the OEG::BIN (TCP) and the DDS::OMD::Snapshot (TCP), SenderCompID, SenderSubID, TargetCompID, TargetSubID for the OEG::FIX (TCP)).

Clients may consider these publicly available source code examples, which allow to connect to both the OEG and DDS:

| Location                  | Description                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>access-client-rust</b> | Rust source code example which implements connection to native binary Trading Port and Market Data (snapshot, stream, and replay). |
| <b>access-client-cpp</b>  | C++ source code example which implements connection to native binary Trading Port and Market Data (snapshot, stream, and replay).  |
| <b>access-client-java</b> | Java source code example which implements connection to native binary Trading Port and Market Data (snapshot, stream, and replay). |

To run any of the three above sources you need to be familiar with the following information:

### 7.2. GPW WATS PROD/PROD-BIS CONNECTIVITY GUIDE

(This section will be defined in the next phase of the project).

## 8. APPENDIX A

### 8.1. EUAT

#### 8.1.1. TRADING PORTS

| Port                    | Parameter     | Description                                                     | Site A                                    |                 | Site B        |               |
|-------------------------|---------------|-----------------------------------------------------------------|-------------------------------------------|-----------------|---------------|---------------|
| <b>OEG::BIN</b>         | IP address    | TCP/IP address of the service.                                  | 91.234.144.196                            | 91.234.144. 197 | 91.234.144.68 | 91.234.144.69 |
|                         | Port          | TCP/IP port of the service.                                     | 10132                                     | 10132           | 10132         | 10132         |
|                         | Connection ID | Numerical identifier of the connection.                         | Individually set with WSE Exchange Member |                 |               |               |
|                         | Token         | Textual secret allowing authorization of the connection.        | Individually set with WSE Exchange Member |                 |               |               |
| <b>OEG::FIX</b>         | IP address    | TCP/IP address of the service.                                  | 91.234.144.196                            | 91.234.144. 197 | 91.234.144.68 | 91.234.144.69 |
|                         | Port          | TCP/IP port of the service.                                     | 10133                                     | 10133           | 10133         | 10133         |
|                         | SenderCompID  | Numerical value used to identify company sending the message.   | Individually set with WSE Exchange Member |                 |               |               |
|                         | TargetCompID  | Numerical value used to identify company receiving the message. | Individually set with WSE Exchange Member |                 |               |               |
|                         | Token         | Textual secret allowing authorization of the connection.        | Individually set with WSE Exchange Member |                 |               |               |
| <b>OEG::DCP<br/>FIX</b> | IP address    | TCP/IP address of the service.                                  | 91.234.144.196                            |                 | 91.234.144.68 |               |
|                         | Port          | TCP/IP port of the service.                                     | 10134                                     |                 | 10134         |               |

#### 8.1.2. MARKET DATA

##### Full Market Data

| Port                                      | Parameter     | Description                             | Site A                                    | Site B        |
|-------------------------------------------|---------------|-----------------------------------------|-------------------------------------------|---------------|
| <b>DDS::OMD::<br/>Snapshot<br/>AllGPW</b> | IP address    | TCP port of the service.                | 91.234.144.198                            | 91.234.144.70 |
|                                           | Port          | TCP/IP port of the service.             | 10150                                     | 10150         |
|                                           | Connection ID | Numerical identifier of the connection. | Individually set with WSE Exchange Member |               |

| Port                             | Parameter     | Description                                              | Site A                                    | Site B        |
|----------------------------------|---------------|----------------------------------------------------------|-------------------------------------------|---------------|
|                                  | Token         | Textual secret allowing authorization of the connection. | Individually set with WSE Exchange Member |               |
| <b>DDS::OMD::Replay AllGPW</b>   | IP address    | TCP/IP address of the service.                           | 91.234.144.198                            | 91.234.144.70 |
|                                  | Port          | TCP/IP port of the service.                              | 10152                                     | 10152         |
| Port                             | Parameter     | Description                                              | Site A                                    | Site B        |
| <b>DDS::OMD::Snapshot NonGPW</b> | IP address    | TCP port of the service.                                 | 91.234.144.199                            | 91.234.144.71 |
|                                  | Port          | TCP/IP port of the service.                              | 10154                                     | 10154         |
|                                  | Connection ID | Numerical identifier of the connection.                  | Individually set with WSE Exchange Member |               |
|                                  | Token         | Textual secret allowing authorization of the connection. | Individually set with WSE Exchange Member |               |
| <b>DDS::OMD::Replay NonGPW</b>   | IP address    | TCP/IP address of the service.                           | 91.234.144.199                            | 91.234.144.71 |
|                                  | Port          | TCP/IP port of the service.                              | 10156                                     | 10156         |

**Best Bid Offer**

| Port                             | Parameter     | Description                                              | Site A                                    | Site B        |
|----------------------------------|---------------|----------------------------------------------------------|-------------------------------------------|---------------|
| <b>DDS::BBO::Snapshot AllGPW</b> | IP address    | TCP port of the service.                                 | 91.234.144.198                            | 91.234.144.70 |
|                                  | Port          | TCP/IP port of the service.                              | 10151                                     | 10151         |
|                                  | Connection ID | Numerical identifier of the connection.                  | Individually set with WSE Exchange Member |               |
|                                  | Token         | Textual secret allowing authorization of the connection. | Individually set with WSE Exchange Member |               |
| <b>DDS::BBO::Replay AllGPW</b>   | IP address    | TCP/IP address of the service.                           | 91.234.144.198                            | 91.234.144.70 |
|                                  | Port          | TCP/IP port of the service.                              | 10153                                     | 10153         |

| Port                             | Parameter     | Description                                              | Site A                                    | Site B        |
|----------------------------------|---------------|----------------------------------------------------------|-------------------------------------------|---------------|
| <b>DDS::BBO::Snapshot NonGPW</b> | IP address    | TCP port of the service.                                 | 91.234.144.199                            | 91.234.144.71 |
|                                  | Port          | TCP/IP port of the service.                              | 10155                                     | 10155         |
|                                  | Connection ID | Numerical identifier of the connection.                  | Individually set with WSE Exchange Member |               |
|                                  | Token         | Textual secret allowing authorization of the connection. | Individually set with WSE Exchange Member |               |

|                                    |            |                                |                |               |
|------------------------------------|------------|--------------------------------|----------------|---------------|
| <b>DDS::BBO::Replay<br/>NonGPW</b> | IP address | TCP/IP address of the service. | 91.234.144.199 | 91.234.144.71 |
|                                    | Port       | TCP/IP port of the service.    | 10157          | 10157         |

#### 8.1.2.1. Multicast Groups

| Site   | Multicast Groups       | IP address | Port  |
|--------|------------------------|------------|-------|
| Site A | #M11 Full Depth AllGPW | 233.1.3.1  | 31001 |
| Site A | #M12 BBO AllGPW        | 233.1.3.2  | 31002 |
| Site A | #M13 Full Depth NonGPW | 233.1.3.3  | 31003 |
| Site A | #M14 BBO NonGPW        | 233.1.3.4  | 31004 |
| Site   | Multicast Groups       | IP address | Port  |
| Site B | #M21 Full Depth AllGPW | 233.1.2.1  | 21001 |
| Site B | #M22 BBO AllGPW        | 233.1.2.2  | 21002 |
| Site B | #M23 Full Depth NonGPW | 233.1.2.3  | 21003 |
| Site B | #M24 BBO NonGPW        | 233.1.2.4  | 21004 |

## 8.2. PRE-PROD (PROD) - WILL BE UPDATED WITH NEXT RELEASE

#### 8.2.1. TRADING PORTS

| Port                | Parameter     | Description                                                                                                 |
|---------------------|---------------|-------------------------------------------------------------------------------------------------------------|
| <b>OEG::BIN</b>     | IP address    | TCP/IP address of the service.                                                                              |
|                     | Port          | TCP/IP port of the service.                                                                                 |
|                     | Connection ID | Numerical identifier of the connection.                                                                     |
|                     | Token         | Textual secret allowing authorization of the connection.                                                    |
| <b>OEG::FIX</b>     | IP address    | TCP/IP address of the service.                                                                              |
|                     | Port          | TCP/IP port of the service.                                                                                 |
|                     | SenderCompID  | Numerical value used to identify company sending the message.                                               |
|                     | TargetCompID  | Numerical value used to identify company receiving the message.                                             |
|                     | TargetSubID   | Numerical value used to identify specific individual or unit intended to receive message.                   |
|                     | Token         | Textual secret allowing authorization of the connection. Transmitted in RawData field of the Logon message. |
| <b>OEG::DCP FIX</b> | IP address    | TCP/IP address of the service.                                                                              |
|                     | Port          | TCP/IP port of the service.                                                                                 |

#### 8.2.2. MARKET DATA

| Port                      | Parameter     | Description                                              |
|---------------------------|---------------|----------------------------------------------------------|
| <b>DDS::OMD::Snapshot</b> | IP address    | TCP/IP address of the service.                           |
|                           | Port          | TCP/IP port of the service.                              |
|                           | Connection ID | Numerical identifier of the connection.                  |
|                           | Token         | Textual secret allowing authorization of the connection. |
| <b>DDS::OMD::Stream</b>   | IP address    | UDP multicast IP address of the service.                 |
|                           | Port          | UDP port of the service.                                 |
| <b>DDS::OMD::Replay</b>   | IP address    | TCP/IP address of the service.                           |
|                           | Port          | TCP/IP port of the service.                              |

## 8.3. PROD BIS - WILL BE UPDATED WITH NEXT RELEASE

### 8.3.1. TRADING PORTS

| Port                | Parameter     | Description                                                                                                 |
|---------------------|---------------|-------------------------------------------------------------------------------------------------------------|
| <b>OEG::BIN</b>     | IP address    | TCP/IP address of the service.                                                                              |
|                     | Port          | TCP/IP port of the service.                                                                                 |
|                     | Connection ID | Numerical identifier of the connection.                                                                     |
|                     | Token         | Textual secret allowing authorization of the connection.                                                    |
| <b>OEG::FIX</b>     | IP address    | TCP/IP address of the service.                                                                              |
|                     | Port          | TCP/IP port of the service.                                                                                 |
|                     | SenderCompID  | Numerical value used to identify company sending the message.                                               |
|                     | SenderSubID   | Numerical value used to identify specific message originator.                                               |
|                     | TargetCompID  | Numerical value used to identify company receiving the message.                                             |
|                     | TargetSubID   | Numerical value used to identify specific individual or unit intended to receive message.                   |
| <b>OEG::DCP FIX</b> | Token         | Textual secret allowing authorization of the connection. Transmitted in RawData field of the Logon message. |
|                     | IP address    | TCP/IP address of the service.                                                                              |
| <b>OEG::DCP FIX</b> | Port          | TCP/IP port of the service.                                                                                 |

### 8.3.2. MARKET DATA

| Port                      | Parameter     | Description                                              |
|---------------------------|---------------|----------------------------------------------------------|
| <b>DDS::OMD::Snapshot</b> | IP address    | TCP/IP address of the service.                           |
|                           | Port          | TCP/IP port of the service.                              |
|                           | Connection ID | Numerical identifier of the connection.                  |
|                           | Token         | Textual secret allowing authorization of the connection. |
| <b>DDS::OMD::Stream</b>   | IP address    | UDP multicast IP address of the service.                 |
|                           | Port          | UDP port of the service.                                 |
| <b>DDS::OMD::Replay</b>   | IP address    | TCP/IP address of the service.                           |
|                           | Port          | TCP/IP port of the service.                              |

## 9. APPENDIX B

Table below represents supported MPLS operators, however this is not a defining list and GPW can support other providers too.

| Operator                            | Unicast support | Multicast support | Supported technology |                                                                 |
|-------------------------------------|-----------------|-------------------|----------------------|-----------------------------------------------------------------|
| Orange Polska S.A                   | YES             | YES               | MPLS/L2              | <a href="https://www.orange.pl/">https://www.orange.pl/</a>     |
| Exatel S.A.                         | YES             | YES               | MPLS/L2              | <a href="https://exatel.pl/">https://exatel.pl/</a>             |
| ATMAN Sp. z o.o.                    | YES             | YES               | MPLS/L2              | <a href="https://www.atman.pl/">https://www.atman.pl/</a>       |
| Colt IQ, Colt Technology Services   | YES             | YES               | MPLS                 | <a href="https://www.colt.net/">https://www.colt.net/</a>       |
| ICE Data, Intercontinental Exchange | YES             | YES               | MPLS                 | <a href="https://www.theice.com/">https://www.theice.com/</a>   |
| Hawe Telekom                        | YES             | YES               | L2                   | <a href="https://hawetelekom.com/">https://hawetelekom.com/</a> |
| T-Mobile                            | YES             | YES               | L2                   | <a href="https://www.t-mobile.pl/">https://www.t-mobile.pl/</a> |
| Netia                               | YES             | YES               | L2                   | <a href="https://www.netia.pl/">https://www.netia.pl/</a>       |